

DIGITAL BOOKSTORE

Final Project Report

A book review web application focused on finance, investing and personal growth.

INT1059 - Advanced Web

Assessment 3 - Final Project (Pair)

Weighting: 40% - Due Module 12

Submitted by

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GitHub Repository:

github.com/Willian-Yudy-F/advanced-web-prototype

1. Executive Summary

The Digital Bookstore is a review-focused web application developed as the final deliverable of INT1059 Advanced Web. It allows visitors to browse a curated collection of finance and personal-development books, filter them by genre, read detailed descriptions, submit star-rated reviews, and manage a personal list of favourite titles.

The project was built in pair by Luiza Miranda Gomes (student ID 15186) and Willian Yudy Futema (student ID 15160) and extends the proposal produced for Assessment 1 and the prototype delivered in Assessment 2. All functional requirements listed in the Assessment Brief 3 have been implemented on a PHP/MySQL stack running locally via the XAMPP environment and are backed by a relational schema with users, books, reviews and favourites tables.

The visual language of the interface is inspired by leading online bookshops such as Booktopia, using a classic forest-green palette with antique-gold accents to reinforce the idea that reading is an investment in personal growth.

2. Project Links

All code, database scripts and documentation required to review and reproduce the project are available at the locations below.

GitHub repository: <https://github.com/Willian-Yudy-F/advanced-web-prototype>

Local URL (after XAMPP setup): <http://localhost/bookstore/>

Submission package: 15186_Gomes_Luiza_INT1059_Assessment_3.zip and 15160_Futema_Willian_INT1059_Assessment_3.zip (same content).

No remote hosting URL is provided because the project is intended to be reviewed locally from the zipped source using XAMPP, as permitted by the brief. The README included in the package explains how to restore the database and run the site in under five minutes.

3. Compliance with the Brief

The table below maps every functional element listed in the Assessment Brief 3 to the file(s) that implement it in this project.

Brief requirement	How it was implemented	Evidence (files)
Home page with random products	Random ORDER BY RAND() LIMIT 8 query feeds a responsive card grid on the landing page with a hero banner.	index.php
Navigation menu with dropdowns	Top navbar with logo, search and account links + secondary bar with a 'Genres' hover dropdown and direct genre tabs.	navbar.php
Category filtering by genre	Six genres (Finance, Investing, Business, Mindset, Lifestyle, Habits) filter via ?genre=... using a prepared	index.php, navbar.php

Brief requirement	How it was implemented	Evidence (files)
	statement.	
Detailed product view (title, synopsis, metadata, review form)	Individual book page shows cover, title, author, genre, synopsis and the review submission form.	book.php
User registration & sign-in	password_hash on registration, password_verify on login, sessions stored in \$_SESSION['user_id'], prepared statements throughout.	register.php, login.php, logout.php
Product rating system (stars)	1-5 star select in the review form; average rating is computed per book and displayed as filled/empty stars on every card.	book.php, index.php
Search functionality	Navbar form posts to search.php with q and genre params; LIKE search across title, author and description with prepared statements.	search.php, navbar.php
User favourites list (add/remove/manage)	Logged users get an 'Add to Favourites' button on every card and a dedicated page to view and remove favourites.	favorites.php, toggle_favorite.php
User account page (manage reviews, update email/password/username)	Account page lists the user's submitted reviews and exposes a form to update name, email and password (re-hashed on save).	account.php

4. Technology Stack

We intentionally kept the stack small so the project is easy to reproduce on any laptop with XAMPP installed and easy for graders to inspect.

Layer	Technology	Why
Server	Apache 2.4 (XAMPP)	Stable local dev environment pre-configured for PHP and MySQL.
Language	PHP 8.2	Module requirement; native mysqli + password_hash available.
Database	MariaDB 10 via	Relational schema fits users-books-reviews-

Layer	Technology	Why
	phpMyAdmin	favourites relations.
Frontend	HTML5 + CSS3 (custom)	Full control over the Booktopia-inspired design; no framework bloat.
Auth	PHP sessions, password_hash/verify	Industry standard, secure by default, taught in class.
Security	mysqli prepared statements, htmlspecialchars	Prevents SQL injection and XSS in forms and listings.
Version control	Git + GitHub	Required by the brief; used for pair collaboration.

5. Database Schema

The relational schema contains four tables. Foreign keys between reviews/favorites and users/books enforce referential integrity, and a unique constraint on (user_id, book_id) in favorites prevents duplicate bookmarks.

Table	Key columns	Purpose
users	id (PK), name, email (unique), password (hashed), created_at	Stores registered users.
books	id (PK), title, author, description, image, genre, created_at	Catalog of reviewable books.
reviews	id (PK), book_id (FK), user_id (FK), rating (1-5), comment, created_at	One review per user per visit to a book page.
favorites	id (PK), user_id (FK), book_id (FK), created_at; UNIQUE(user_id, book_id)	User's bookmarked books.

The full CREATE TABLE and seed data are exported to books.sql and users.sql in the project folder. The database is named advanced_web in MariaDB.

6. Team Contribution

The project was developed in pair. Both members contributed to design decisions, implementation, testing and documentation. The table below summarises the primary responsibility of each member, although work was reviewed and merged collaboratively via GitHub.

Area	Luiza Miranda Gomes	Willian Yudy Futema
Proposal & planning (Assessment 1)	Theme research, user stories, wireframes.	Requirements mapping, scope control.
Prototype (Assessment 2)	Home page and catalog layout.	Database modelling, initial CRUD.
Database (Assessment 3)	Reviews and favourites schema, seed data.	Users schema, constraints, SQL export.
Backend PHP	Auth, favourites, review submission.	Search, book detail, account management.
Frontend & UX	CSS theme, cards, hero, responsive design.	Navbar, dropdowns, accessibility review.
Testing & documentation	Functional test pass, README, this report.	Browser testing, README, this report.

7. Setup and Deployment

7.1 Prerequisites

- XAMPP 8.2 or later installed (Apache + MySQL/MariaDB + phpMyAdmin).
- A modern browser (Chrome, Firefox, Safari or Edge).
- Around 50 MB of free disk space for book cover images.

7.2 Steps

- **Unzip** 15186_Gomes_Luiza_INT1059_Assessment_3.zip and copy the bookstore folder into /Applications/XAMPP/xamppfiles/htdocs/ (macOS) or C:/xampp/htdocs/ (Windows).
- **Start Apache and MySQL** from the XAMPP control panel.
- **Open phpMyAdmin** (<http://localhost/phpmyadmin>) and create a database named advanced_web (utf8mb4).
- **Import** books.sql and users.sql (both in the bookstore folder) using the Import tab.
- **Open** <http://localhost/bookstore/> to access the site.
- **Create a new account** via the Create account link to test the registration, login, review and favourites flows.

7.3 Default credentials

A demo account is included for convenience: email demo@bookstore.com / password demo1234. If the credentials do not work, simply create a new account via the Create account page.

8. Testing

Every functional requirement listed in the brief has been manually tested on Chrome and Safari on macOS. The matrix below is the final smoke-test log executed before submission.

Feature	Steps to reproduce	Result
Home random products	Open / -> see 8 random cards. Reload: order changes.	Pass
Genre filter	Click FINANCE in the subnav -> only finance books appear.	Pass
Search	Type 'money' in the navbar search -> related books listed.	Pass
Register	Create a new account with unique email -> redirected to login.	Pass
Login	Sign in with the new credentials -> Favourites + My Account links show.	Pass
Review submission	Open a book page, pick 4 stars, write comment, submit.	Pass
Favourites	Click Add to Favourites -> appears under Favourites page. Remove works.	Pass
Account page	My Account -> edit email/password/username -> update persists.	Pass
Logout	Click Logout -> session cleared, account links become Sign in/Create account.	Pass

9. Design Decisions and Reflection

9.1 Visual language

The interface takes cues from Booktopia with a few deliberate deviations. Instead of Booktopia's bright navy/orange, we chose a forest-green and antique-gold palette that echoes classic book-binding colours and communicates a more academic, serious mood appropriate for finance and personal-growth titles. Cards use a subtle hover lift, restrained shadows and a consistent 3:4 cover aspect ratio to keep the grid calm.

9.2 Security

User passwords are never stored in plain text: we use PHP's `password_hash()` with the default Argon2/Bcrypt algorithm on registration and `password_verify()` on login. Every database read that depends on user input uses a prepared statement via `mysqli_prepare` + `mysqli_stmt_bind_param`,

protecting against SQL injection. All dynamic content printed into HTML passes through `htmlspecialchars()` to block XSS.

9.3 Challenges

- The session had to be started before any include that emitted HTML, otherwise the initial version of `login.php` triggered an HTTP 500 because of the headers-already-sent state. Moving `session_start()` to the very top of the file solved it.
- Book cover filenames sometimes had mixed case and accidental whitespace; we standardised them to lowercase `.jpg` and moved every file to the `/images` folder so `file_exists` checks work on the case-sensitive macOS filesystem.
- Balancing the visual richness we wanted (hero, dropdowns, starred cards) with the simplicity expected from a single-lecturer review required keeping the CSS in a single file and reusing a single `.book-card` style everywhere.

9.4 What we learned

Working in pair forced us to write clearer commit messages and to keep features behind small, well-named files rather than one large controller. Implementing favourites and reviews as separate tables (rather than columns on users) made the account page and the home page a lot easier to assemble because we could write focused SQL for each use case. Finally, we gained first-hand appreciation for why frameworks introduce helpers like ORM and templating engines: most of the bugs we fixed were because of duplicated `mysqli` code or manually echoed HTML.

9.5 Future improvements

- Deploy the site to a free PHP host (InfinityFree or 000webhost) and include the live URL next to the GitHub link.
- Add pagination to the search results and on the home page when the catalog grows beyond 50 books.
- Allow users to upload their own avatar and make the reviews section show the reviewer's name.
- Introduce a CSRF token on all POST forms.

10. References

PHP Group 2024, PHP 8.2 Manual, The PHP Group, viewed 19 April 2026, <<https://www.php.net/manual/en/>>.

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Mozilla Developer Network 2026, MDN Web Docs: CSS Grid Layout, viewed 15 April 2026, <https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_grid_layout>.

Apache Friends 2026, XAMPP Installers and Documentation, viewed 12 April 2026, <<https://www.apachefriends.org/>>.